ModelIt!

Student Activity Pack

Ecosystem Engineers

How One Species Creates a World

Grade NE | NGSS 5-LS2-1, 5-ESS2-1, 3-5-ETS1-3 | Systems and System Models

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **In your Level 1 lesson, you modeled beavers, trees, dam size, and pond size. What do you think comes AFTER the pond?**

2. **How many different animals do you think could live in a beaver pond? List as many as you can.**

3. **What would happen to all those animals if the beavers disappeared?**

4. **Draw what you think a beaver-created wetland looks like from above.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Beaver Population | Tree Availability | Rainfall | Dam Size | Water Level | Wetland Area | Biodiversity

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Thriving Colony

Full resources available

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Drought Year

Low rainfall despite beavers

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Beavers Removed

Zero beavers

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Water Temperature — The temperature of the pond — affected by depth, shade, and flow rate

Sediment Filtering — How much dirt and pollution the wetland traps — clean water supports more life

Predator Population — Number of predators like otters, eagles, and herons that move into the rich ecosystem

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does a family of beavers create an entire ecosystem that hundreds of species depend on?
Use evidence from your simulation to support your answers.

1. **How is the Level 2 model different from the Level 1 model? What does adding more components reveal?**

2. **Trace the cascade: if Tree Availability drops by half, what happens at each step all the way to Biodiversity?**

3. **Why are beavers sometimes called 'nature's water managers'?**

4. **What surprised you most about the simulation results?**

5. **How could this model help real conservation planners decide where to reintroduce beavers?**

8 STEM Challenge: Wetland Watercolor Mural

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YOUR ENGINEERING MISSION

After running the simulation, create a large collaborative mural showing the ecosystem at three stages: no beavers, small colony, and thriving colony. Every species depicted must match what the model predicted.

THE SCENARIO

A nature center wants a mural that teaches visitors how beavers create wetlands. Your team will use model data to ensure every species and habitat feature shown is scientifically accurate at each stage.

Engineering Design Thinking

1. HOW WILL YOU DIVIDE THE MURAL INTO THREE PANELS (NO BEAVERS, SMALL COLONY, THRIVING)?

2. WHICH SPECIES WILL YOU INCLUDE IN EACH PANEL AND WHY?

3. HOW WILL YOU SHOW THE WATER LEVEL DIFFERENCE BETWEEN STAGES?

4. HOW WILL YOU MAKE THE MURAL BOTH SCIENTIFICALLY ACCURATE AND VISUALLY COMPELLING?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Wetland Ecologists study how wetland ecosystems function and design restoration projects. They work with government agencies, conservation organizations, and universities. They earn $55,000–$90,000/year.